Rodrigo Pizarro Alegría

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About me

Mathematical Engineer who builds machine-learning systems end to end, from raw data to production inference, monitoring, and iteration. My thesis and deep-learning work center on computer vision: convolutional and Kolmogorov-Arnold networks (KAN) for image classification and transfer learning, with calibration and predictive uncertainty treated as first-class concerns. In industry I ship production predictive ML models at the scale of 1-2M+ users on cloud infrastructure, and build the full-stack software around them.

Skills

Computer Vision & Deep Learning: Image classification, convolutional neural networks, Kolmogorov-Arnold networks, transfer learning, feature representations, model calibration, uncertainty quantification. OpenCV, PyTorch.
Machine Learning: XGBoost / LightGBM, scikit-learn, gradient boosting, probabilistic and hurdle models, survival / hazard models, time-series forecasting, RAG / LLMs (LangChain).
Software Development: Python, C/C++, SQL, FastAPI, REST APIs, Docker, Git, React / Next.js, TypeScript, Supabase, PostgreSQL.
Cloud & Data: Google Cloud Platform: Vertex AI, BigQuery, Cloud Storage.
Mathematics: Probability, statistics, optimization, linear algebra, stochastic processes, functional and complex analysis.
Languages: Spanish (native), English (professional working proficiency).

Experience

Data Scientist, Tenpo (Fintech, Credicorp group) — Santiago, Chile — December 2025 – Present

- Own an end-to-end production ML pipeline forecasting Customer Lifetime Value at panel scale (1-2M+ users): data handling in BigQuery, modular Python training on Vertex AI, deployment, backtesting, and monitoring. Decomposed operating margin into six revenue and cost components and trained direct multi-horizon XGBoost models.
- Built a competing-risks discrete-time hazard model predicting credit-card closure (churn) at short and long horizons, by cause (delinquency vs. voluntary), across a 1-2M+ user panel. Validated out-of-time with rolling-origin backtests.
- Built two-branch hurdle models (classifier + Gamma regression) for zero-inflated financial targets with per-horizon probability recalibration, and modeled portfolio credit-risk exposure validated against actuarial benchmark proxies.

Mathematical Modeling Engineer, Demafront SPA — Valparaíso, Chile — August 2024 – December 2025

- Built and shipped a cloud-native SaaS forecasting API in Python from scratch (ML model over hundreds of thousands of time series), improving accuracy metrics by 20%+ in production.
- Designed a novel hierarchical forecast-reconciliation method for product trees with thousands of nodes where standard MinT (SoTA technique) is ill-posed or unfeasible; 10%+ accuracy gain over MinT and bottom-up baselines across retail clients.

GenAI Researcher, Laureate Inc. — Remote (Contractor) — March 2024 – August 2024

- Built and evaluated retrieval-augmented generation (RAG) pipelines for academic assistant agents (LangChain, Ragas, TruLens); improved faithfulness, answer relevance, and context precision over baseline.

Teaching Assistant, USM – Valparaíso, Chile — March 2020 – August 2024

- **Mathematics:** Differential, integral, and multivariable calculus, vector calculus, ordinary and partial differential equations, complex analysis.
- **Physics:** Thermodynamics, fluid mechanics, mechanical waves.
- **Computer Science:** Algorithms, computational complexity.

Projects

Transfer Learning in Image Classification with Dempster–Shafer Layers — Engineering thesis (100/100)

- Designed and trained convolutional and Kolmogorov–Arnold architectures from scratch, with evidential (belief-function) output heads, for image classification under distribution shift where source and target label distributions diverge.
- Benchmarked evidential heads against softmax and temperature-scaled baselines on predictive uncertainty and calibration; studied set-valued outputs enabling cautious classification and abstention. Advisor: Dr. Alejandro Veloz.

Véndelo (in development) — social-commerce mobile app (short-video peer-to-peer marketplace)

- Built full-stack with Next.js / React (TypeScript) and Supabase/Postgres schema, authentication, media storage. Solo, end to end: product, backend, frontend, and deployment.
- Ran a full production-readiness audit ahead of mobile launch: fixed stale refresh-token handling and a build-breaking SSR bug, and prepared the iOS release.

Education

Universidad Técnica Federico Santa María — Civil Mathematical Engineer December 2024
Thesis: Transfer Learning in Image Classification Using Dempster–Shafer Layers. Grade: 100/100.
Universidad Técnica Federico Santa María — Bachelor's Degree in Engineering Sciences December 2023

Certifications

Computer Vision — OpenCV BootCamp
Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms — Stanford Online

Awards & Honors

Bronze Medal, CMAT 2019 — third place in the top individual category of a national mathematics championship for high-school students, organized by Chilean universities.
USM Excellence List (2020) and **USM Academic Merit Award (2022)** — ranked among the top two students by GPA in the 2019 cohort.